Question 1 [10 Marks]

A retail company is analyzing historical demand data to improve its short-term forecasting accuracy for inventory planning. The management wants to evaluate different averaging techniques and assess forecast accuracy before making operational decisions.

The recorded demand over the last 8 periods is given below:

Period	Demand
1	50
2	55
3	53
4	60
5	58
6	62
7	65
8	63

Tasks:

a) Calculate the **3-period moving average forecast** for periods 4 to 9. [3]

b) Calculate the **weighted moving average forecast for period 9** using weights: [4]

- **0.5 (most recent)**
- **0.3**
- **0.2**

c) Using the forecasts obtained in part (a), evaluate the forecasting performance by computing: [3]

- Mean Absolute Deviation (MAD)
- Mean Squared Error (MSE)

Question 2 [10 Marks]

A technology company is testing different forecasting models to predict future demand for its digital service. The company wants to compare the responsiveness and accuracy of exponential smoothing models with different smoothing constants.

The demand data is as follows:

Period	Demand
1	42
2	40
3	45
4	43
5	47
6	44
7	48

Assume the initial forecast: $F_2 = A_1 = 42$

Tasks:

a) Using **Exponential Smoothing**, compute forecasts for periods 3 to 7 using: [4]

- $\alpha = 0.2$
- $\alpha = 0.5$

b) For both values of α, calculate: [4]

- Forecast errors, Absolute errors and Mean Absolute Deviation (MAD)

c) Based on your results: [2]

- Identify which value of $\boldsymbol{\alpha}$ provides better forecasting performance
- Briefly justify your answer in terms of responsiveness and error